CO4_AT1_DATA INTERPRETATION

1. Real-Time Detection System
A surveillance system uses the following techniques:

Method	Accuracy	Speed
Viola-Jones	78%	Fast
YOLO	92%	Moderate
Deep Learning	95%	Slow

Compare the three techniques and evaluate which method should be selected for a surveillance system operating under strict latency constraints while maintaining acceptable accuracy. Justify your decision considering deployment conditions.

2. Robust Object Detection

Method	Normal	Occlusion	Low Light
Viola-Jones	80%	60%	55%
YOLO	90%	75%	70%
Deep Learning	95%	85%	80%

Compare the robustness of the three methods under normal, occluded, and low-light conditions. Evaluate which method provides the best overall performance and discuss the trade-offs involved.

3. Optical Flow Analysis
Two optical flow methods are available:

- Method A: Produces stable motion vectors but misses small motions.
- Method B: Detects fine motion but produces noisy vectors.

Compare the two methods and evaluate which method is more suitable for motion tracking in dynamic scenes. Justify your answer based on accuracy, stability, noise sensitivity, and application requirements.

4. Model Generalization and Overfitting – 10 Marks

Model	Training Accuracy	Testing Accuracy
A	98%	70%
B	90%	88%

Compare the two models and evaluate which model is more reliable for real-world deployment. Explain the generalization ability and overfitting risk of each model and justify your selection.

5. System Performance under Constraints

Model	Accuracy	Latency	Cost
A	88%	50 ms	Low
B	92%	80 ms	Moderate
C	95%	120 ms	High

Compare the three models and evaluate which model provides the best balance between accuracy, latency, and cost for a real-time application. Justify your choice considering practical deployment constraints.

ANSWERS

1. Real-Time Detection System

The surveillance system uses three object-detection techniques: Viola-Jones, YOLO, and Deep Learning. They differ in accuracy and processing speed.

Method	Accuracy	Speed	Evaluation
Viola-Jones	78%	Fast	Low accuracy but very quick
YOLO	92%	Moderate	High accuracy with reasonable speed
Deep Learning	95%	Slow	Highest accuracy but slower

Viola-Jones:

Viola-Jones is a fast object-detection technique. Its major advantage is its low computational requirement and fast response. However, its accuracy is only 78%, which may result in missed detections or false detections. Therefore, it may not be suitable when high detection reliability is required.

YOLO:

YOLO provides 92% accuracy with moderate processing speed. It is capable of detecting objects efficiently and is widely suitable for real-time applications. Although it is slower than Viola-Jones, the improvement in accuracy is significant.

Deep Learning:

Deep Learning provides the highest accuracy of 95%. This means it can identify objects more reliably. However, its slow processing speed may introduce latency. It may also require more powerful hardware and computational resources.

Evaluation

For a surveillance system, both accuracy and speed are important. A system with very high accuracy but slow response may not be useful for real-time monitoring. Similarly, a very fast system with poor accuracy may miss important objects.

YOLO offers a good compromise between the two requirements.

Conclusion

YOLO should be selected because it provides 92% accuracy with moderate speed. It offers a practical balance between detection accuracy and real-time response. Deep Learning can be selected when accuracy is the highest priority and sufficient computing resources are available.

Final choice: YOLO.

2. Robust Object Detection

The performance of the three methods is tested under normal, occlusion, and low-light conditions.

Method	Normal	Occlusion	Low Light
Viola-Jones	80%	60%	55%
YOLO	90%	75%	70%
Deep Learning	95%	85%	80%

Normal Conditions

Under normal conditions:

- Viola-Jones achieves 80%.
- YOLO achieves 90%.
- Deep Learning achieves 95%.

Deep Learning provides the highest performance.

Occlusion Conditions:
When objects are partially hidden:

- Viola-Jones = 60%
- YOLO = 75%
- Deep Learning = 85%

Deep Learning performs better because it can extract more complex features and recognize partially visible objects.

Low-Light Conditions:
Under low-light conditions:

- Viola-Jones = 55%
- YOLO = 70%
- Deep Learning = 80%

Again, Deep Learning provides the best result.

Robustness Analysis
The decrease in accuracy from normal to low-light conditions is:

- Viola-Jones: 25 percentage points
- YOLO: 20 percentage points
- Deep Learning: 15 percentage points

Therefore, Deep Learning experiences the smallest decrease and is the most robust method.

Trade-Offs
The major advantage of Deep Learning is its high accuracy and robustness. However, it may require:

- Higher computational power
- More memory
- Greater processing time

- More complex implementation

YOLO provides a good middle ground, while Viola-Jones is advantageous when computational resources are limited.

Conclusion

Deep Learning is the best method for robust object detection because it achieves the highest accuracy in all three conditions and maintains better performance when conditions become difficult.

Final choice: Deep Learning.

3. Optical Flow Analysis

Optical flow is used to estimate the movement of objects between consecutive frames of a video.

Two methods are available:

Method A

Method A:

- Produces stable motion vectors.
- Provides consistent movement information.
- Has low noise.
- Is reliable for general motion tracking.
- May miss small or subtle movements.

Method B

Method B:

- Detects fine and small movements.
- Has higher sensitivity.
- Can capture detailed motion.
- Produces noisy motion vectors.
- Noise can make tracking less stable.

Comparison

The selection depends on the application requirement.

If the application requires stable and reliable tracking, Method A is better. For example, tracking a moving vehicle or person generally requires consistent motion information.

If the application requires detection of very small movements, Method B may be more suitable. However, the noisy vectors may require additional filtering or smoothing.

Evaluation

For a general dynamic-scene tracking application, stable motion estimation is important because noisy vectors can cause:

- Incorrect movement estimation
- Unstable object tracking
- False motion detection
- Reduced tracking accuracy

Method A avoids these problems to a greater extent.

Conclusion

Method A is more suitable for general motion tracking because it produces stable and reliable motion vectors. Method B should be selected only when detecting fine movements is more important than motion stability.

Final choice: Method A.

4. Model Generalization and Overfitting

The two models have the following performance:

Model	Training Accuracy	Testing Accuracy	Difference
A	98%	70%	28%
B	90%	88%	2%

Model A Analysis

Model A achieves 98% training accuracy, which appears excellent. However, its testing accuracy is only 70%.

The difference is:

98% − 70% = 28 percentage points

This is a large gap between training and testing performance. It indicates that Model A has likely overfitted the training data.

The model has learned the training examples very well but has difficulty handling new or unseen data.

Model B Analysis

Model B has:

- Training accuracy = 90%
- Testing accuracy = 88%

The difference is:

90% − 88% = 2 percentage points

This is a very small gap. Therefore, Model B has better generalization ability.

Generalization

Generalization refers to the ability of a machine-learning model to perform well on new and unseen data.

For real-world deployment, generalization is more important than achieving extremely high training accuracy.

Comparison

Model A has higher training accuracy but poor testing performance. Model B has slightly lower training accuracy but much better testing performance.

Therefore:

Model A → High overfitting risk

Model B → Low overfitting risk

Conclusion

Model B should be selected for real-world deployment because its training and testing accuracies are close. It demonstrates better generalization and is more likely to provide reliable predictions on unseen data.

Final choice: Model B.

5. System Performance Under Constraints

The models are compared using accuracy, latency, and cost.

Model	Accuracy	Latency	Cost
A	88%	50 ms	Low
B	92%	80 ms	Moderate
C	95%	120 ms	High

Model A

Model A has:

- 88% accuracy
- 50 ms latency
- Low cost

Its major advantage is its fast response and low deployment cost. However, its accuracy is lower than the other two models.

It is suitable for applications where speed and cost are more important than maximum accuracy.

Model B

Model B provides:

- 92% accuracy
- 80 ms latency
- Moderate cost

It provides a significant accuracy improvement over Model A while maintaining reasonable latency.

Therefore, Model B provides a balanced solution.

Model C

Model C achieves the highest accuracy:

95%

However, it has:

- 120 ms latency
- High cost

The higher latency may affect real-time response. Its high cost may also make large-scale deployment difficult.

Practical Deployment Analysis

For a real-time system, latency should be reasonably low. The system should provide accurate results without taking too much time to respond.

Model A is fastest but sacrifices accuracy.

Model C is most accurate but has high latency and cost.

Model B lies between the two and provides a better overall balance.

Conclusion

Model B is the most suitable choice for practical real-time deployment. Its 92% accuracy is sufficiently high, its 80 ms latency is reasonable, and its moderate cost makes deployment more practical.

Final choice: Model B.

Overall Comparison

Question	Best Choice	Main Reason
1. Real-Time Detection	YOLO	Best accuracy-speed balance
2. Robust Detection	Deep Learning	Best performance in difficult conditions
3. Optical Flow	Method A	Stable and less noisy motion vectors

Question	Best Choice	Main Reason
4. Generalization	Model B	Low overfitting and good testing accuracy
5. System Constraints	Model B	Best balance of accuracy, latency and cost